

Notes: Mas and Moretti (AER, 2009)

Peers at Work

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1 Big picture and incremental contribution

1.1 Big picture

- **Question.** Do peer effects exist in ordinary workplaces, and if so, are they driven by free-riding, social pressure, prosocial preferences, or some other mechanism?
- **Setting.** The paper studies cashiers in a large supermarket chain. The production process is group-like because customers choose among checkout lines, so one cashier's speed affects coworkers' workload.
- **Data advantage.** The authors use scanner data to measure each cashier's productivity at very high frequency. Productivity is items scanned per second in ten-minute cells.
- **Main result.** A positive shock to coworkers' permanent productivity raises the focal worker's productivity. The baseline coefficient is about 0.15: a 10 percent increase in coworker productivity raises individual productivity by about 1.5 percent.
- **Heterogeneity.** Low-productivity workers respond strongly to high-productivity peers; high-productivity workers are not meaningfully slowed down by low-productivity peers.
- **Mechanism.** The spillover appears only when the productive coworker can observe the focal worker, and it is stronger when the two workers have frequently overlapped in the past. This supports social pressure rather than altruism or pure knowledge spillovers.
- **Managerial implication.** Optimal scheduling may increase skill diversity within shifts, because mixing high- and low-productivity workers raises low-productivity workers' effort without reducing high-productivity workers' effort.

1.2 Incremental contribution of this paper

- **Field evidence on workplace peer effects.** Prior work had strong theory and lab evidence on peer pressure, team incentives, and cooperation, but relatively little field evidence from naturally occurring low-skill workplaces. This paper brings high-frequency field evidence to the question.
- **High-frequency individual productivity measurement.** The paper improves on studies using aggregate plant- or firm-level productivity by constructing individual productivity at ten-minute frequency from scanner transactions.
- **Mechanism identification.** The paper goes beyond estimating an average spillover. It uses register orientation and line of sight to distinguish social pressure from prosocial preferences and knowledge spillovers.
- **Peer-effect timing.** The dynamic event-study-style estimates show that productivity responds exactly when coworker composition changes, helping rule out pre-existing demand shocks.
- **Workplace design implication.** The paper links peer effects to scheduling and wage-setting. A high-productivity worker's value includes both own output and induced productivity in coworkers.
- **Organizational economics contribution.** The paper demonstrates that informal social incentives can partially substitute for formal performance pay when individual output is hard to contract on.

2 Data and measurement

2.1 Supermarket setting and sample

The paper studies cashiers in six stores of a national supermarket chain. The data cover a two-year period and contain detailed scanner records. The unit of observation in most regressions is a cashier-ten-minute cell.

- Transactions include the number of items scanned and transaction length.
- Productivity is measured as items scanned per second.
- A ten-minute interval is included only if the cashier transacts in that interval.
- Managers and assistant managers are excluded because they may enter registers in response to demand spikes.
- The baseline sample in Table 2 has roughly 1.7 million cashier–ten-minute observations.

Interpretation: The setting is attractive because there is a clearly measured individual output even though the workflow contains group externalities. One slow cashier pushes customers toward other cashiers, so productivity choices impose costs on coworkers.

2.2 Permanent productivity and coworker productivity

For cashier i , permanent productivity is estimated from a first-stage model with individual fixed effects. Let y_{itcs} denote log productivity of worker i in time period t , day c , and store s . The authors first estimate a model of the form

$$y_{itcs} = \theta_i + M' \mathcal{P}_{Ci} + \pi N_{tcs} + \tau R_{ics} + \gamma_{tcs} + e_{itcs},$$

where θ_i is individual permanent productivity, N_{tcs} is the number of checkers on duty, R_{ics} captures register location, and γ_{tcs} are hour-of-day by day-of-week by store controls.

The average permanent productivity of coworkers in a shift is

$$\bar{\theta}_{-itcs} = \frac{1}{n_{tcs} - 1} \sum_{j \neq i} \theta_j.$$

Interpretation: Permanent productivity is intended to measure worker ability or usual speed, not the worker’s effort in the current interval. This lets the paper ask whether exogenous-looking changes in the ability composition of coworkers affect current effort.

2.3 Spatial orientation and overlap measures

The paper uses register positions to classify coworkers into two sets:

- **Observing set:** coworkers who face the focal worker and therefore can observe the focal worker’s effort.
- **Observable set:** coworkers whom the focal worker can see.

It also uses previous schedule overlap to proxy for repeated interactions. Coworkers are classified into low, medium, and high overlap depending on how frequently their past schedules overlapped.

Interpretation: These variables are what allow the paper to open the black box. Social pressure predicts that workers respond when they are observed. Prosocial preferences predict that workers respond when they observe a faster peer, regardless of whether that peer observes them.

3 Model and empirical specifications

3.1 Conceptual framework: free-riding versus peer pressure

Without social concerns, a more productive coworker can make the focal worker free-ride. If worker effort is costly and the employer observes total output imperfectly, the focal worker bears the full effort cost but

receives only part of the benefit from higher total productivity. In the paper’s theoretical discussion, when coworker j becomes more productive, optimal effort by worker i falls absent social pressure:

$$\frac{de_i^*}{d\theta_j} < 0.$$

With peer pressure, worker i ’s objective includes a peer-pressure cost:

$$\max_{e_i} Q_i U(w) - C_i(e_i) - P(e_i, e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_N).$$

Interpretation: The sign of the empirical peer effect is ambiguous ex ante. Free-riding predicts a negative response to highly productive coworkers. Peer pressure predicts a positive response, especially when the focal worker’s low effort is visible to peers.

3.2 Baseline estimating equation

The baseline model is obtained by first-differencing productivity:

$$\Delta y_{itcs} = \alpha + \beta \Delta \bar{\theta}_{-itcs} + \pi \Delta N_{tcs} + e_{itcs}.$$

Here Δy_{itcs} is the change in focal-worker log productivity between adjacent ten-minute periods, and $\Delta \bar{\theta}_{-itcs}$ is the change in the average permanent productivity of coworkers over the same interval.

Interpretation: The coefficient β is the key spillover effect. Positive β means the focal worker works faster when the coworker composition becomes more productive.

3.3 Dynamic response model

To test timing and pre-trends, the paper estimates a model with leads and lags of coworker productivity shocks:

$$\Delta y_{itcs} = \alpha + \sum_{k=-7}^7 \beta_k \Delta \bar{\theta}_{-i,t+k,cs} + \pi \Delta N_{tcs} + e_{itcs}.$$

Interpretation: If high-productivity workers are scheduled in anticipation of demand increases, lead coefficients should predict productivity before the peer shock. Figure 2 shows the response occurs at the shock date, which supports the identifying assumption.

3.4 Heterogeneity by worker ability

The paper tests whether low-ability and high-ability workers respond differently:

$$\Delta y_{itcs} = \alpha + \beta \Delta \bar{\theta}_{-itcs} + \lambda \Delta \bar{\theta}_{-itcs} L_i + \pi \Delta N_{tcs} + e_{itcs},$$

where $L_i = 1$ if worker i has above-average permanent productivity.

Interpretation: The interaction identifies whether high-productivity workers are less responsive to coworkers. A negative λ means low-productivity workers account for most of the peer effect.

3.5 Spatial-orientation model

Let G denote the observing set and E denote the observable set. The paper estimates separate coefficients for peer shocks in these sets:

$$\Delta y_{itcs} = \alpha + \beta^G \Delta \bar{\theta}_{-itcs}^G + \beta^E \Delta \bar{\theta}_{-itcs}^E + \pi \Delta N_{tcs} + e_{itcs}.$$

Interpretation: Social pressure predicts $\beta^G > 0$ and $\beta^E \approx 0$. Prosocial behavior predicts the opposite pattern. The estimates strongly support the social-pressure prediction.

3.6 Previous-overlap model

The paper further interacts peer productivity shocks with low, medium, and high previous schedule overlap:

$$\Delta y_{itcs} = \alpha + \beta^L \Delta \bar{\theta}_{-itcs}^L + \beta^M \Delta \bar{\theta}_{-itcs}^M + \beta^H \Delta \bar{\theta}_{-itcs}^H + \pi \Delta N_{tcs} + e_{itcs}.$$

Interpretation: Social pressure should be stronger when repeated interaction creates scope for reputational sanctions, shame, or future reciprocity. Table 7 supports this prediction.

4 Identification strategy

4.1 Core source of variation

Identification comes from within-worker, within-day changes in coworker composition over ten-minute windows. Worker shifts do not perfectly overlap, so the set of coworkers changes as workers enter and exit registers.

- Permanent productivity is estimated at the worker level.
- The empirical shock is a change in the average permanent productivity of coworkers.
- First differencing removes fixed worker traits and many persistent store/day differences.
- Controls include changes in the number of workers on duty.

Interpretation: The design asks whether a given cashier speeds up when the composition of coworkers around that cashier becomes more productive, holding fixed short-run changes in staffing levels.

4.2 Main identifying assumption

The main identifying assumption is that ten-minute changes in coworker permanent productivity are uncorrelated with unobserved shocks to the focal worker’s productivity, conditional on staffing changes and controls.

Potential threats include:

- management scheduling high-productivity workers exactly when demand is about to rise;
- high-productivity workers entering or leaving in response to demand shocks;
- shared resources such as baggers creating mechanical relationships between worker composition and productivity;
- spatial sorting of workers to particular registers.

4.3 Validation tests

The paper addresses these threats using several strategies:

- **Dynamic response tests:** Figure 2 shows no anticipation before the peer shock.
- **Predictable demand tests:** Table 4 shows that changes in predicted sales do not differentially predict entry or exit of high-productivity personnel in a way that can explain the main result.
- **Lagged actual demand tests:** Table 5 shows that lagged changes in actual sales volume do not generate the required sorting pattern.
- **Personnel-change tests:** Figure 4 shows little relation between net personnel changes and changes in average permanent productivity.
- **Spatial-orientation tests:** Table 6 and Figure 5 show that spillovers operate when workers are observed, not merely when they observe faster workers.
- **Repeated-interaction tests:** Table 7 shows stronger effects when coworkers have higher previous schedule overlap.

Interpretation: The identification is strongest when viewed as a package. The baseline estimates show peer effects; the timing, sales-volume, spatial, and overlap tests rule out several major alternatives and identify social pressure as the mechanism.

5 Tables and figures

5.1 Table 1 and Figure 1: Descriptive Statistics and Distribution of Workers' Permanent Productivity

Read:

- Table 1 reports store-level descriptive statistics across six stores and all stores.
- Average productivity is about 0.17 items per second in ten-minute intervals.
- The sample averages about 6.9 checkers on duty in a ten-minute interval.
- Figure 1 shows the distribution of estimated permanent productivity for 394 workers.
- The fixed-effect distribution is centered at zero by construction and has substantial dispersion, which creates variation in coworker quality.

Interpretation: The table and figure establish that the empirical setting has enough worker-level productivity dispersion to study peer effects. Without meaningful variation in permanent productivity, coworker composition shocks would be too small to identify spillovers.

Economic message: Permanent productivity heterogeneity is the input into the peer-effect design: the paper converts scheduling variation into variation in coworker ability exposure.

TABLE 1—DESCRIPTIVE STATISTICS

	Store # 1 (1)	Store # 2 (2)	Store # 3 (3)	Store # 4 (4)	Store # 5 (5)	Store # 6 (6)	All stores (7)
Share of ten-minute interval that checkers are transacting	0.67 [0.32]	0.61 [0.25]	0.64 [0.28]	0.69 [0.26]	0.68 [0.24]	0.60 [0.26]	0.65 [0.27]
Minutes per customer	1.4 [1.0]	1.2 [1.1]	1.6 [1.1]	1.3 [1.1]	1.4 [0.86]	1.4 [0.91]	1.4 [1.0]
Productivity in ten-minute intervals	0.18 [0.09]	0.16 [0.07]	0.17 [0.08]	0.16 [0.07]	0.18 [0.07]	0.20 [0.08]	0.17 [0.08]
Checkers on duty in ten- minute intervals	5.8 [1.9]	5.9 [1.6]	4.7 [1.7]	7.7 [2.1]	8.3 [2.4]	7.0 [2.3]	6.9 [2.4]
Estimated individual fixed effects	[0.07]	[0.12]	[0.08]	[0.08]	[0.09]	[0.09]	[0.09]
Average coworker permanent productivity	[0.04]	[0.06]	[0.04]	[0.03]	[0.04]	[0.04]	[0.04]
Change in coworker permanent productivity	[0.02]	[0.03]	[0.03]	[0.02]	[0.02]	[0.02]	[0.02]

Notes: Main entries are means. Figures in brackets are standard deviations. The estimated fixed effects are mean 0 by construction. The units of observation are checker $\times$ ten-minute cells. Individual productivity is defined as the number of items scanned per seconds spent in transactions over a ten-minute period. Specifically, for each worker on the shift, we sum the number of items that a worker scanned over a ten-minute period. We divide this number by the total number of seconds that the worker was in a transaction, where a transaction is defined as the time between when the first item is scanned and when the payment is completed and the receipt for the transaction is given to the customer. We include in our definition of productivity only periods when transactions are occurring. The sample excludes any observations that do not occur in the 7 AM–8 PM interval. The sample excludes transactions involving managers. Average coworker permanent productivity is the average permanent productivity of all workers in a shift, excluding permanent productivity of the focal worker i .

(a) Table 1: Descriptive statistics.

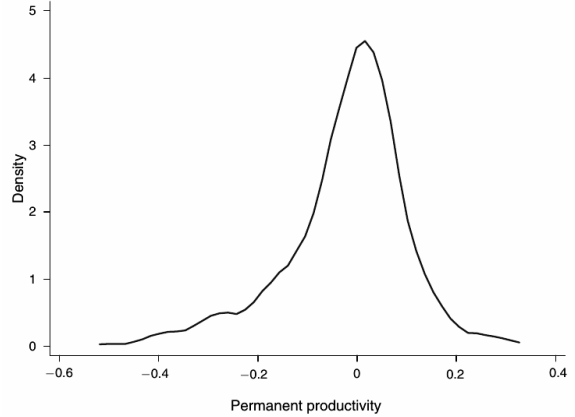


FIGURE 1. DISTRIBUTION OF WORKERS' PERMANENT PRODUCTIVITY

Figure 1 shows a kernel density estimate of estimated worker permanent productivity (the parameter θ_i in equation (7)). The sample is 394 workers. We used an Epanechnikov kernel and a bandwidth of 0.05.

(b) Figure 1: Distribution of workers' permanent productivity.

5.2 Table 2 and Figure 2: Baseline Peer Effects and Dynamic Response

Read:

- Table 2 estimates the baseline coefficient on the change in average coworker permanent productivity.
- Columns 1 and 2 estimate coefficients of 0.15, implying that a 10 percent increase in coworker productivity raises focal productivity by about 1.5 percent.
- Column 4 shows the response is primarily to positive coworker-productivity shocks.
- Column 5 shows that entry of above-average workers raises focal-worker productivity and exit lowers it.

- Figure 2 shows that the productivity response occurs exactly when the shock arrives and then persists.

Interpretation: Table 2 is the main average-effect result. Figure 2 is the main timing validation. Together, they show that the spillover is positive, immediate, and not driven by trends before high-productivity coworkers enter.

Economic message: In this workplace, peer effects are strong enough to offset free-riding incentives. A high-productivity worker contributes directly and indirectly by raising the effort of coworkers.

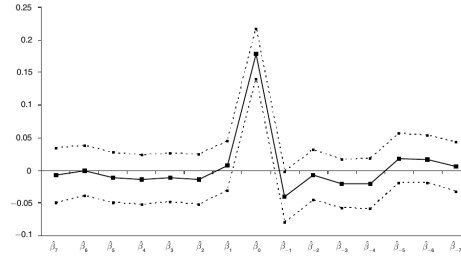
TABLE 2—THE EFFECT OF CHANGES IN AVERAGE COWORKER PERMANENT PRODUCTIVITY ON FOCAL WORKER PRODUCTIVITY
(Dependent variable is the difference in log productivity of the focal worker between t and $t - 1$)

	(1)	(2)	(3)	(4)	(5)
Δ Average coworker permanent productivity	0.15 (0.02)	0.15 (0.02)	0.13 (0.03)	-0.03 (0.03)	
Δ Average coworker permanent productivity $\times$ positive Δ indicator				0.24 (0.05)	
Positive Δ indicator				0.004 (0.001)	
Entry of above average productivity worker					0.010 (0.001)
Exit of an above average productivity worker					-0.005 (0.001)
Observations	1,718,052	1,718,052	823,274	1,718,052	1,732,941
Additional controls?		Yes			
No net change in number of workers from $t - 1$ to t ?			Yes		

Notes: OLS estimates. Bootstrapped standard errors that are robust to serial correlation are in parentheses. The parametric bootstrap procedure is described in Section III and in the Appendix. Observations are checker $\times$ ten-minute cells. See Table 1 notes for additional details on the sample. The symbol Δ indicates the difference in the relevant variable between periods t and $t - 1$. The dependent variable is the change in the log productivity of a checker across consecutive ten-minute periods. Individual productivity is defined as the number of items scanned per second transacting over a ten-minute period. Average coworker permanent productivity is computed as the simple average of coworker permanent productivity components (estimated by fitting equation (7)) in a given shift, where the average excludes focal worker i . All specifications other than (3) control for the change in the number of workers on duty across consecutive ten-minute periods. In column 3 the sample consists of time periods where there was no change in the number of workers on duty. The additional controls in column 2 are dummies for store by hour of day and by day of week and a quadratic polynomial in the number of ten-minute periods a worker has been on duty. In column 5 we also include in the model dummies whether there was any entry or exit into or out of the shift, irrespective of the productivity of the worker entering or departing (not reported). "Entry" is defined as one if in a given day a checker was on duty at time t but not on duty at time $t - 1$, and zero otherwise. "Exit" is defined as one if in a given day a checker was on duty at time t but not on duty at time $t + 1$, and zero otherwise. "Positive Δ indicator" is one if " Δ average coworker permanent productivity" is positive and zero otherwise.

(a) Table 2: Effect of changes in average coworker permanent productivity.

Panel A: Estimated coefficients on the lag and lead terms for changes in the average permanent productivity of coworkers, where the dependent variable is change in focal worker productivity. See equation (8).



Panel B: Cumulative response.

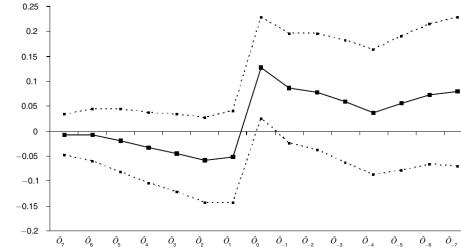


FIGURE 2: DYNAMIC RESPONSE FOLLOWING A SHOCK TO THE AVERAGE PERMANENT PRODUCTIVITY OF COWORKERS

Notes: Figures based on equation (8) in the text. In panel A, we plot the estimated coefficients on the lagged and lead terms in (8) along with accompanying 95 percent confidence intervals (dotted line). For example, β_7 is the coefficient on the seventh lag term. In panel B we plot the cumulative response: $\hat{\alpha}_n = \hat{\beta}_7 + \hat{\beta}_6 + \dots + \hat{\beta}_n$, for $n = -7$ through 7. For example, $\hat{\alpha}_{-2} = \hat{\beta}_7 + \hat{\beta}_6 + \hat{\beta}_5 + \hat{\beta}_4 + \hat{\beta}_3 + \hat{\beta}_2 + \hat{\beta}_1 + \hat{\beta}_0$. In estimating (8), standard errors were computed using the bootstrapped procedure described in Section III and in the Appendix.

(b) Figure 2: Dynamic response following a coworker-productivity shock.

5.3 Table 3 and Figure 3: Heterogeneity in Responsiveness by Ability Type

Read:

- Table 3 shows that low-productivity workers account for most of the peer effect.
- The coefficient for below-average workers is about 0.24.
- The interaction with above-average-worker status is about -0.19 , leaving a small positive effect for high-productivity workers.
- Figure 3 estimates worker-specific spillovers and plots them against permanent productivity.
- The fitted relationship slopes downward: low-productivity workers are most responsive to peer pressure.

Interpretation: These results explain why worker mixing matters. Peer effects raise low-productivity workers' effort without making high-productivity workers much less productive.

Economic message: Skill diversity can be productive in group settings because high-ability workers create positive pressure on low-ability workers without being dragged down symmetrically.

TABLE 3—HETEROGENEITY IN RESPONSIVENESS BY ABILITY TYPE
(Dependent variable is the difference in log productivity of the focal worker between t and $t - 1$)

	(1)	(2)
Δ Average coworker permanent productivity	0.24 (0.03)	0.24 (0.03)
Δ Coworker permanent productivity × above average worker	−0.19 (0.04)	−0.18 (0.04)
Observations	1,718,052	1,718,052
Additional controls?		Yes

Notes: Specifications present estimates from OLS models. The sample is the same as in Table 2, column 1. See Table 2 notes for more details on sample. Bootstrapped standard errors that are robust to serial correlation are in parentheses. The parametric bootstrap procedure is described in Section III and in the Appendix. The units of observation are checker × ten-minute cells. The symbol Δ indicates the difference in the relevant variable between periods t and $t - 1$. The dependent variable is the change in the log productivity of a checker across ten-minute periods. Individual productivity is defined as the number of items scanned per second transacting over a ten-minute period. “Above average worker” refers to a worker with above average permanent productivity in his or her store. Average coworker permanent productivity is computed as the simple average of worker permanent productivity components (estimated by fitting equation (7)) in a given shift, where the average excludes the focal worker i . The additional controls in column 2 are dummies for store by hour of day and by day of week and a quadratic polynomial in the number of ten-minute periods a worker has been on duty.

(a) Table 3: Heterogeneity in responsiveness by ability type.

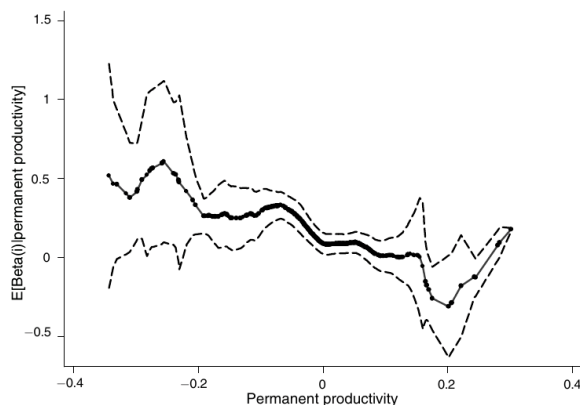


FIGURE 3. RELATIONSHIP BETWEEN INDIVIDUAL PERMANENT PRODUCTIVITY AND WORKER-SPECIFIC SPILLOVER EFFECT

is figure plots the local-linear regression fit for the relationship between the estimated worker specific spillover effect (the parameter β_i in equation (10)) against that worker's permanent productivity (the parameter θ_i). The dashed lines are the 95 percent confidence intervals. We use an Epanechnikov kernel with a bandwidth of 0.05. The expression is weighted by inverse variances of the estimated worker-specific spillover effects. The

(b) Figure 3: Spillover effect by permanent productivity.

5.4 Tables 4 and 5: Sales Volume, Entry, and Exit Tests

Read:

- Table 4 tests whether predicted demand changes are associated with entry or exit of high-productivity workers.
- High-productivity workers are not more likely to enter when predicted demand is rising; if anything, the evidence goes against the confounding story.
- Table 5 uses lagged actual sales changes instead of predicted sales.
- Again, the required sorting pattern is not present.

Interpretation: These tables are identifying-assumption tests. If managers inserted fast workers exactly when demand was rising, the baseline peer effect could be spurious. The tables show little evidence of that behavior.

Economic message: The estimated peer effects are unlikely to be simple demand-management effects. They reflect coworker interactions rather than endogenous staff scheduling around customer volume.

TABLE 4—RELATIONSHIP BETWEEN CHANGES IN PREDICTABLE SALES VOLUME AND EXIT AND ENTRY PROBABILITIES OF PERSONNEL
(Linear probability models)

	Dependent variable:	
	Entry at t (1)	Exit at t (2)
Δ Predicted log average transactions between $t - 1$ and t	0.029 (0.048)	-0.086 (0.009)
Permanent productivity	0.090 (0.007)	0.077 (0.007)
Δ Predicted log average transactions between $t - 1$ and $t \times$ permanent productivity	-0.078 (0.048)	0.21 (0.07)
Observations	2,029,970	2,029,970

Notes: The units of observation are checker $\times$ ten-minute cells. "Exit at t " means that in a given day the checker is last observed working in period t . "Entry at t " means that in a given day the checker is first observed in period t . Bootstrapped standard errors that are robust to serial correlation are in parentheses. The parametric bootstrap procedure is described in Section III and in the Appendix. Column 1 corresponds to equation (11) in the text and column 2 corresponds to equation (12). The symbol Δ indicates the difference in the relevant variable between periods t and $t - 1$. To compute predicted transactions for worker i , we compute the average number of items scanned in store $\times$ day of week $\times$ time of day cells, excluding any observations where worker i was on duty (to prevent a mechanical correlation between a worker's permanent productivity and predicted demand). See text for additional details.

(a) Table 4: Predictable sales volume and personnel entry/exit.

TABLE 5—RELATIONSHIP BETWEEN LAGGED CHANGES IN SALES VOLUME AND ENTRY AND EXIT PROBABILITIES OF PERSONNEL
(Linear probability models)

	Dependent variable	
	Entry at t (1)	Exit at t (2)
Δ log items sold between $t - 2$ and $t - 1$	-0.032 (0.0002)	0.006 (0.0007)
Permanent productivity	0.090 (0.005)	0.081 (0.005)
Δ log items sold between $t - 2$ and $t - 1 \times$ permanent productivity	-0.054 (0.008)	-0.0005 (0.007)
Observations	2,050,558	2,050,558

Notes: "Exit at t " means that in a given day the checker is last observed working in period t . "Entry at t " means that in a given day the checker is first observed in period t . Bootstrapped standard errors that are robust to serial correlation are in parentheses. The parametric bootstrap procedure is described in Section IIIA and in the Appendix. The units of observation are checker $\times$ ten-minute cells. The symbol Δ indicates the difference in the relevant variable between periods t and $t - 1$.

(b) Table 5: Lagged sales volume and personnel entry/exit.

5.5 Figure 4 and Figure 5: Net Personnel Changes and Spatial Orientation Dynamic Response

Read:

- Figure 4 shows box plots by store relating net changes in personnel on duty to changes in average permanent productivity.
- There is little systematic relationship; the median change in average permanent productivity is small relative to its standard deviation.
- Figure 5 breaks dynamic responses into observing and observable sets.
- The response appears when the focal worker is in the line of sight of high-productivity coworkers, not when the focal worker merely sees high-productivity coworkers.

Interpretation: Figure 4 addresses staffing-composition confounds. Figure 5 addresses the mechanism. The spatial pattern is exactly what social pressure predicts and not what prosocial preferences predict.

Economic message: Workers increase effort when their underperformance can be observed by faster coworkers. Visibility is the key incentive channel.

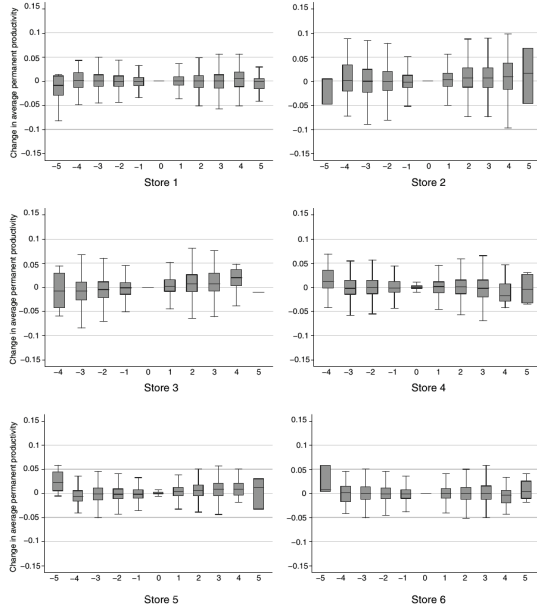
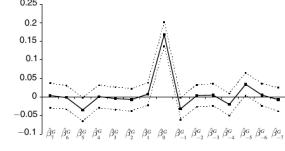


FIGURE 4. RELATIONSHIP BETWEEN NET CHANGES IN PERSONNEL ON DUTY AND CHANGES IN THE AVERAGE PERMANENT PRODUCTIVITY OF WORKERS

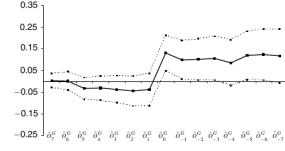
Notes: The x-axis is the net change in the number of personnel across consecutive ten-minute periods. Absolute values in changes larger than five are excluded due to small sample sizes. The y-axis is the corresponding change in the average permanent productivity of workers on duty. The units of observation are store $\times$ ten-minute period cells. Change in

(a) Figure 4: Net personnel changes and changes in average permanent productivity.

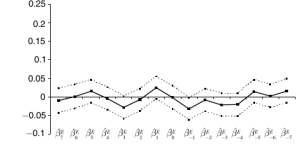
Panel A: Dynamic response to a positive average coworker permanent productivity shock in the observing set



Panel B: Cumulative response to a positive average coworker permanent productivity shock in the observing set



Panel C: Dynamic response to a positive average coworker permanent productivity shock in the observable set



Panel D: Cumulative response to a positive average coworker permanent productivity shock in the observable set

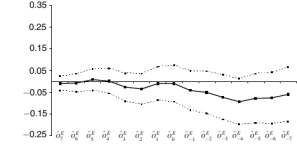


FIGURE 5. DYNAMIC RESPONSE MODELS BROKEN DOWN BY SPATIAL ORIENTATION

Note: In panel A we plot the estimated coefficients on the lagged and lead terms from estimating (14) along with accompanying 95 percent confidence intervals (dotted line). For example, $\hat{\beta}_7$ is the coefficient on the seventh lead term. In panel B we plot the cumulative response: $\hat{\sigma} = \hat{\beta}_7 + \hat{\beta}_6 + \dots + \hat{\beta}_u$ for $u = -7$ through 7. Panels C and D correspond to the estimated lag and lead coefficients and the cumulative response for equation (15), respectively. Standard errors were computed using the bootstrapped procedure described in Section III and in the Appendix.

In panels A and B of Figure 5, we consider the focal worker's dynamic response to a change in

(b) Figure 5: Dynamic response by spatial orientation.

5.6 Tables 6 and 7: Spatial Orientation, Proximity, and Previous Scheduling Overlap

Read:

- Table 6 shows that average coworker permanent productivity in the observing set raises focal productivity, while productivity in the observable set does not.
- The effect is stronger for coworkers who are closer in the observing set.
- Table 7 shows the peer effect is small for low-overlap coworkers and larger for medium/high-overlap coworkers.
- Presence effects also follow the same pattern: entering or exiting coworkers matter when they are in the observing set and have had prior overlap with the focal worker.

Interpretation: These are the main mechanism tables. Social pressure requires both observability and a relational environment in which future interactions or sanctions matter. The estimates line up with both requirements.

Economic message: Informal incentives are relational. Monitoring by unknown or rarely encountered peers is less powerful than monitoring by coworkers with whom one repeatedly interacts.

TABLE 6—EFFECT OF CHANGES OF AVERAGE COWORKER PERMANENT PRODUCTIVITY ON FOCAL WORKER'S PRODUCTIVITY
(Models by spatial orientation and proximity; dependent variable is the difference in log productivity of the focal worker between t and $t - 1$)

	(1)	(2)	(3)	(4)	(5)	(6)
Δ Avg. coworker permanent productivity in observing set	0.17 (0.02)	0.18 (0.02)				
Δ Avg. coworker permanent productivity in observable set	0.01 (0.02)	0.00 (0.02)				
Δ Avg. coworker permanent productivity observing set & closer					0.11 (0.01)	
Δ Avg. coworker permanent productivity in observable set & closer					0.01 (0.01)	
Δ Avg. coworker permanent productivity in observing set & farther					0.06 (0.01)	
Δ Avg. coworker permanent productivity in observable set & farther					0.01 (0.01)	
Δ Presence of worker in observing set			0.031 (0.003)	0.031 (0.004)		
Δ Presence of worker in observable set			-0.03 (0.003)	-0.022 (0.004)		
Δ Presence of worker in observing set & closer					0.027 (0.002)	
Δ Presence of worker in observable set & closer					-0.026 (0.002)	
Δ Presence of worker in observing set & farther					0.013 (0.002)	
Δ Presence of worker in observable set & farther					0.008 (0.002)	
Using only middle registers?		Yes		Yes		
Observations	1,649,916	683,933	1,732,941	721,176	1,487,086	1,732,941

Notes: Specifications present estimates from OLS models. Column 1 corresponds to estimating equation (13) in the text. The remaining columns are variants. The original sample is the same as the one used in Table 2. The sample sizes differ from those in Table 2 because, for some individuals, there are no observations in the relevant subgroup (e.g., average coworker permanent productivity in the observable set). Bootstrapped standard errors robust to serial correlation are in parentheses. The parametric bootstrap procedure is described in Section IIIA and in the Appendix. The units of observation are checker $\times$ ten-minute cells. The dependent variable is the change in the log productivity of a checker across ten-minute periods. "Observing set" refers to workers who are facing worker i . "Observable set" refers to workers whom i is facing. "Closer" refers to workers who are one or two positions away from i . "Farther" denotes workers who are three or four positions away from i . The permanent productivity averages are taken over the indicated subgroups. The symbol Δ indicates the difference in the relevant variable between periods t and $t - 1$. For example, " Δ avg. coworker permanent productivity in the observable set & closer" denotes the change in the average permanent

TABLE 7—THE EFFECT OF CHANGES OF AVERAGE COWORKER PERMANENT PRODUCTIVITY ON FOCAL WORKER'S PRODUCTIVITY
(Models by previous scheduling overlap between coworkers and spatial orientation; dependent variable is the difference in log productivity of the focal worker between t and $t - 1$)

	(1)	(2)	(3)
(I) Δ Avg. coworker permanent productivity: low overlap	0.01 (0.01)	(I) Δ Avg. coworker permanent productivity: observing set & low overlap	0.01 (0.02)
(II) Δ Avg. coworker permanent productivity: medium overlap	0.06 (0.01)	(II) Δ Avg. coworker permanent productivity: observing set & medium overlap	0.11 (0.02)
(III) Δ Avg. coworker permanent productivity: high overlap	0.07 (0.02)	(III) Δ Avg. coworker permanent productivity: observing set & high overlap	0.09 (0.03)
		(IV) Δ Avg. coworker permanent productivity: observable set & low overlap	-0.02 (0.02)
		(V) Δ Avg. coworker permanent productivity: observable set & medium overlap	0.01 (0.02)
		(VI) Δ Avg. coworker permanent productivity: observable set & high overlap	-0.06 (0.03)
p -value: H_0 (I) = (II)	0.002	p -value: H_0 (I) = (II)	0.001
H_0 (I) = (III)	0.001	H_0 (I) = (III)	0.010
H_0 (II) = (III)	0.673	H_0 (II) = (III)	0.710
		H_0 (IV) = (V)	0.266
		H_0 (IV) = (VI)	0.234
		H_0 (V) = (VI)	0.049
Observations	1,658,491		1,658,491

Notes: Each column refers to a different OLS regression. Bootstrapped standard errors that are robust to serial correlation are in parentheses. The parametric bootstrap procedure is described in Section III and in the Appendix. The sample is based on the one used in Table 2. The units of observation are checker $\times$ ten-minute cells. The sample sizes differ from those in Table 2 because for some individuals there are no observations in the relevant subgroup (e.g., average coworker permanent productivity in the observable set). Also, the first 31 days of the sample in each store are dropped because we require an initial period to estimate the degree of schedule overlap among workers. The dependent variable is the change in the log productivity of a checker across ten-minute periods. In the case of worker i , "low overlap" denotes the set of workers who have previously overlapped between 0 and 5 percent of i 's schedule. "Medium overlap" denotes workers who have overlapped between 5 and 20 percent of i 's schedule. "High overlap" denotes workers who have overlapped between 20 and 100 percent of i 's schedule. "Observing set" refers to workers facing worker i . "Observable set" refers to workers whom i is facing. The permanent productivity averages are taken over the indicated subgroups. The symbol Δ indicates the difference in the relevant variable between periods t and $t - 1$. For example, variable II in column 2 is the change in the average productivity of coworkers who are in the focal worker's observing set and have medium previous overlap with worker i . In this example, if there are no workers who are positioned in worker i 's observing set (i.e., no workers who are directly looking at i) who have medium previous overlap with i in $t - 1$ and t , this change is coded as zero. If there are no workers positioned in worker i 's observing set in just t or $t - 1$, then this

(a) Table 6: Spatial orientation and proximity.

(b) Table 7: Previous scheduling overlap and spatial orientation.

5.7 Appendix Table 1: Estimated Coefficients from Dynamic Response Models

Read:

- Appendix Table 1 reports the coefficients that are plotted in Figure 2 and Figure 5.
- Column 1 contains the dynamic response coefficients from equation (8).
- Columns 2 and 3 contain the observing-set and observable-set dynamic response coefficients from equations (14) and (15).

Interpretation: The appendix table gives the numerical values behind the key dynamic figures. It is useful for checking that the visual spikes at $t = 0$ and the absence of pretrends are not created by graph scaling.

APPENDIX TABLE 1—ESTIMATED COEFFICIENTS FROM DYNAMIC RESPONSE MODELS

Estimated coefficients from equation (8)			Estimated coefficients from equation (14)			Estimated coefficients from equation (15)		
(1)			(2)			(3)		
$\hat{\beta}_7$	−0.008	(0.021)	$\hat{\beta}_7^G$	0.003	(0.017)	$\hat{\beta}_7^E$	−0.010	(0.017)
$\hat{\beta}_6$	0.000	(0.020)	$\hat{\beta}_6^G$	−0.002	(0.016)	$\hat{\beta}_6^E$	0.001	(0.016)
$\hat{\beta}_5$	−0.012	(0.020)	$\hat{\beta}_5^G$	−0.034	(0.016)	$\hat{\beta}_5^E$	0.015	(0.016)
$\hat{\beta}_4$	−0.014	(0.019)	$\hat{\beta}_4^G$	0.001	(0.015)	$\hat{\beta}_4^E$	−0.005	(0.016)
$\hat{\beta}_3$	−0.011	(0.019)	$\hat{\beta}_3^G$	−0.005	(0.016)	$\hat{\beta}_3^E$	−0.029	(0.016)
$\hat{\beta}_2$	−0.014	(0.020)	$\hat{\beta}_2^G$	−0.008	(0.016)	$\hat{\beta}_2^E$	−0.008	(0.015)
$\hat{\beta}_1$	0.007	(0.019)	$\hat{\beta}_1^G$	0.007	(0.015)	$\hat{\beta}_1^E$	0.025	(0.016)
$\hat{\beta}_0$	0.178	(0.019)	$\hat{\beta}_0^G$	0.168	(0.017)	$\hat{\beta}_0^E$	−0.001	(0.016)
$\hat{\beta}_{-1}$	−0.041	(0.020)	$\hat{\beta}_{-1}^G$	−0.032	(0.015)	$\hat{\beta}_{-1}^E$	−0.032	(0.015)
$\hat{\beta}_{-2}$	−0.007	(0.019)	$\hat{\beta}_{-2}^G$	0.003	(0.015)	$\hat{\beta}_{-2}^E$	−0.009	(0.016)
$\hat{\beta}_{-3}$	−0.020	(0.019)	$\hat{\beta}_{-3}^G$	0.005	(0.015)	$\hat{\beta}_{-3}^E$	−0.021	(0.016)
$\hat{\beta}_{-4}$	−0.021	(0.020)	$\hat{\beta}_{-4}^G$	−0.021	(0.015)	$\hat{\beta}_{-4}^E$	−0.021	(0.016)
$\hat{\beta}_{-5}$	0.018	(0.019)	$\hat{\beta}_{-5}^G$	0.034	(0.015)	$\hat{\beta}_{-5}^E$	0.014	(0.016)
$\hat{\beta}_{-6}$	0.017	(0.019)	$\hat{\beta}_{-6}^G$	0.005	(0.015)	$\hat{\beta}_{-6}^E$	0.003	(0.016)
$\hat{\beta}_{-7}$	0.006	(0.019)	$\hat{\beta}_{-7}^G$	−0.007	(0.016)	$\hat{\beta}_{-7}^E$	0.016	(0.017)

Notes: Coefficients in column 1 are plotted in Figure 2. Coefficients in columns 2 and 3 are plotted in Figure 5. See notes to Figures 2 and 5 for further details.

Appendix Table 1: Estimated coefficients from dynamic response models.

6 Interpretive synthesis

- **Average peer effect.** Productive coworkers increase focal worker effort, contrary to pure free-riding predictions.
- **Heterogeneous response.** Low-productivity workers respond most. High-productivity workers are not hurt by low-productivity peers.
- **Timing.** Responses occur exactly when coworker composition changes, which supports a causal interpretation.
- **Mechanism.** Visibility and repeated interaction are the core channels. This supports social pressure over prosocial preferences or knowledge spillovers.
- **Organizational design.** The optimal shift composition may maximize skill diversity, not skill homogeneity.
- **Incentive design.** Peer effects imply that the marginal product of a high-productivity worker includes induced output from coworkers, so compensation and assignment policies should account for spillovers.

7 Short summary

- The paper studies supermarket cashiers using scanner data at ten-minute frequency.
- Productivity is measured as items scanned per second.
- Permanent productivity is estimated using worker fixed effects.
- The main regressor is the change in average permanent productivity of coworkers working in the same store at the same time.
- The baseline spillover coefficient is about 0.15, implying a 1.5 percent productivity response to a 10 percent coworker-productivity increase.
- Low-productivity workers are the most responsive to high-productivity peers.
- Dynamic estimates show no pre-trends and a sharp response at coworker entry/exit.
- Sales-volume and personnel-change tests reduce concern about endogenous scheduling.

- Spatial-orientation evidence shows that workers respond when they are observed, not merely when they observe others.
- Previous schedule overlap strengthens the effect, consistent with repeated-interaction social pressure.
- The paper concludes that social pressure can partially internalize free-riding externalities in workplaces.

8 Conclusion

8.1 Core conclusion

Mas and Moretti show that peer effects in a real workplace are positive, economically meaningful, and driven primarily by social pressure. Cashiers speed up when high-productivity coworkers enter their shift, especially when those coworkers can observe them and have interacted with them before.

8.2 Economic intuition

In a supermarket, slow workers impose costs on coworkers because customers shift across checkout lines. Formal monetary incentives are weak, but informal peer monitoring creates pressure to do one's fair share. Workers respond to that pressure when their behavior is visible and repeated interaction makes reputation or sanctions plausible.

8.3 Incremental contribution in one sentence

The paper's distinctive contribution is to combine high-frequency individual productivity data, within-shift coworker-composition variation, spatial line-of-sight information, and repeated-overlap measures to identify not only workplace peer effects but also the social-pressure mechanism behind them.

8.4 Takeaway

Workplace productivity depends not only on individual ability and formal incentives but also on social structure. Visibility, repeated interaction, and worker mixing can shape effort in economically important ways.